

From association to mechanism

An evidence standard for sparse features implicated in value-guided choice.

value state → feature activity → model choice

Question: what evidence distinguishes this chain from scaffolding, content tracking, or post-choice activation?

Research methods brief

Association is compatible with four causal stories

The same signed activation gap can support very different interpretations.

FOUR HYPOTHESES

01	Scaffold	option order → feature activity	The feature follows formatting or response position.
02	Content	option wording → feature activity	The feature recognizes what one option says.
03	Consequence	model choice → feature activity	The feature appears after the decision is fixed.
04	Mechanism	value state → feature → choice	The feature participates before or during choice.

WHAT SEPARATES THEM

- 1 **Order**

Does the sign survive option reversal?
- 2 **Timing**

Is evidence present before the answer?
- 3 **Intervention**

Does ablation change choice predictably?

Until then: candidate, not mechanism.

Each alternative explanation needs its own control

A single aggregate score cannot reveal which causal story survived.

ALTERNATIVE	MATCHED COMPARISON	FAILURE PATTERN	INTERPRETATION
Scaffold	Reverse option order; preserve semantic side	Effect follows position	Formatting-sensitive
Content	Compare relevant and unrelated option spans	Effect stays local to wording	Content tracker
Consequence	Move measurement before the answer site	Evidence appears only afterward	Decision readout
Mechanism	Ablate or patch with negative controls	Choice changes in predicted direction	Causal contributor

Reporting rule preserve disagreements across controls instead of collapsing them into one confidence score.

Use the strongest label the evidence earns

The vocabulary should become more specific only as the design becomes more demanding.

DESCRIPTIVE

Associated feature

Signed activation differs across choices.

CLAIM CEILING

association

LOCALIZED

Decision-relevant candidate

Direction survives controls at a relevant site or time.

CLAIM CEILING

candidate involvement

CAUSAL

Mechanistic contributor

Matched intervention changes choice as predicted.

CLAIM CEILING

causal contribution

Mixed evidence is still a result; state exactly where the claim stops.